Math 241, F14 **Name:** Answer

Quiz 2 **SSN:** xxx-xx-

<u>**Instructions:**</u> **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.**

Problem 1. (6 points) Given two vectors $\mathbf{a} =< 3, -2, 6 >$ and $\mathbf{b} =< 4, 0, 3 >$.

(a) Find the angle between $\mathbf{a}$ and $\mathbf{b}$.

$$|a| = \sqrt{3^2+(-2)^2+6^2} = \sqrt{49} = 7$$

$$|b| = \sqrt{4^2+0^2+3^2} = \sqrt{25} = 5$$

$$a \cdot b = 3\cdot 4 + (-2)\cdot 0 + 6\times 3 = 30$$

$$\Rightarrow \cos\theta = \frac{a\cdot b}{|a||b|} = \frac{30}{5\times 7} = \frac{30}{35} = \frac{6}{7} \Rightarrow \theta = \cos^{-1}\left(\frac{6}{7}\right)$$

(b) Find the vector projection of $\mathbf{b}$ onto $\mathbf{a}$, $\text{proj}_{\mathbf{a}}\mathbf{b}$.

$$\text{Proj}_a b = \left(\frac{a\cdot b}{|a|^2}\right) a$$

$$= \left(\frac{30}{7^2}\right) \langle 3, -2, 6 \rangle$$

$$= \left< \frac{90}{49}, -\frac{60}{49}, \frac{180}{49} \right>$$

Problem 2. (4 points) Compute the cross product $\mathbf{a} \times \mathbf{b}$ where $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j} + 7\mathbf{k}$ and $\mathbf{b} = \mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$.

$$a \times b = \begin{vmatrix} i & j & k \\ 2 & -3 & 7 \\ 1 & 4 & -2 \end{vmatrix} = i \begin{vmatrix} -3 & 7 \\ 4 & -2 \end{vmatrix} - j \begin{vmatrix} 2 & 7 \\ 1 & -2 \end{vmatrix} + k \begin{vmatrix} 2 & -3 \\ 1 & 4 \end{vmatrix}$$

$$= i(6-28) - j(-4-7) + k(8-(-3))$$

$$= -22i + 11j + 11k.$$